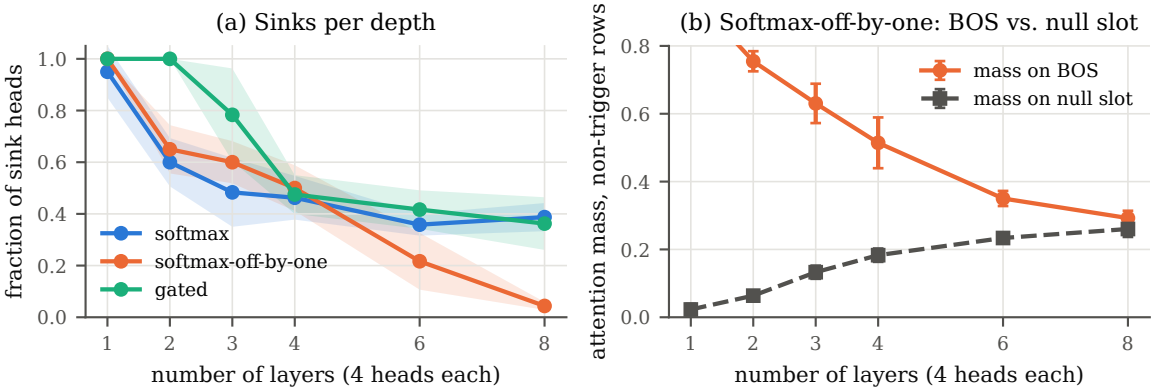
(a) Sinks per depth
fraction of sink heads
1.0
0.8
0.6
0.4
0.2
0.0
softmax
softmax-off-by-one
gated
1
2
3
4
6
8
number of layers (4 heads each)
(b) Softmax-off-by-one: BOS vs. null slot
attention mass, non-trigger rows
0.8
0.6
0.4
0.2
0.0
mass on BOS
mass on null slot
1
2
3
4
6
8
number of layers (4 heads each)